

1.2 REST APIs — Still Needed

Gamma API — Market Discovery & ID Mapping

Base URL: <https://gamma-api.polymarket.com>

Endpoint	Method	Key parameters	Purpose
https://gamma-api.polymarket.com/markets	GET	active=true, volume_num_min, tagId	Discover active markets, get condition_id + token_id for each
<a href="https://gamma-api.polymarket.com/markets/<condition_id>">https://gamma-api.polymarket.com/markets/<condition_id>	GET	condition_id (path param)	Get full metadata for a specific market including resolution criteria

Critical — this must run first. Build a lookup table: condition_id → token_id. Your WebSocket subscriptions use token_id; every Data API call uses condition_id. Without this map you cannot link the two.

Run on startup and re-scan periodically (every 15–30 min) to pick up newly created markets.

Data API — Trades Endpoint

Base URL: <https://data-api.polymarket.com>

Endpoint	Method	Key parameters	Purpose
https://data-api.polymarket.com/trades	GET	market=<condition_id>, limit, before (pagination)	Executed trades: price, size, timestamp. Used for Kyle's lambda, VPIN, and Lee-Ready direction inference

Metrics derived:

- Kyle's lambda — regress price changes on net order flow: $\lambda = \text{cov}(\Delta P, \text{net_order_flow}) / \text{var}(\text{net_order_flow})$
- VPIN — bucket trades into equal volume increments, compute fraction of imbalanced flow per bucket
- Lee-Ready classification — infer buy/sell direction: trade above midpoint = buy, below = sell (needed because Polymarket does not always tag direction)

Polling frequency: Fetch new trades every 100ms. Recalculate lambda and VPIN on each batch.

Data API — Positions Endpoint

Endpoint	Method	Key parameters	Purpose
https://data-api.polymarket.com/positions	GET	market=<condition_id>	Wallet-level position sizes. Used to calculate HHI trader concentration

Metric derived:

- $HHI = \sum(s^2 \text{ for } s \text{ in } [\text{position/total for each wallet}]) \rightarrow 0 = \text{fully distributed}, 1 = \text{one whale controls everything}$

Polling frequency: Every 5 minutes. Positions change slowly.

⚠ One trader using multiple wallets appears as separate participants. HHI will underestimate true concentration — treat high HHI as a strong signal but low HHI as only moderate reassurance.

Data API — Price History Endpoint

Endpoint	Method	Key parameters	Purpose
https://data-api.polymarket.com/prices-history	GET	market=<condition_id>, interval (1h/1d), fidelity	OHLC price history. Used for Hurst exponent, GARCH volatility, jump persistence

Metrics derived:

- Hurst exponent — via Rescaled Range (R/S) analysis. $H < 0.5$ = mean-reverting (noise), $H > 0.5$ = trending (signal). Use as confidence multiplier.
- GARCH volatility — estimates conditional volatility. Detects whether current vol is elevated vs historical baseline.
- Jump persistence — did a sudden price move sustain or reverse? Sustained = fundamental shift. Reversed = emotional overreaction.

Polling frequency: Every 5 minutes. These are slow-moving statistical properties.